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For correspondence:
davidrobert.grimes@tcd.ie

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Determining fragility and robustness to missing data in binary outcome meta-analyses, illustrated with conflicting associations between vitamin D and cancer mortality

David Robert Grimes[1,2]

[1]TCD Biostatistics Unit, Discipline of Public Health and Primary Care, School of Medicine, Trinity College Dublin, Dublin, Ireland • [2]Retraction Watch, The Center For Scientific Integrity, New York, United States

eLife Assessment

This manuscript makes a **valuable** contribution to the concept of fragility of meta-analyses via the so-called 'ellipse of insignificance for meta-analyses' (EOIMETA). The strength of evidence is **convincing**, supported primarily by an example of the fragility of meta-analyses in the association between Vitamin D supplementation and cancer mortality, but the approach could be applied in other meta-analytic contexts. The significance of the work could be enhanced with a more thorough assessment of the impact of between-study heterogeneity, additional case studies, and improved contextualization of the proposed approach in relation to other methods.

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Abstract

Meta-analysis is a vital component in clinical decision making, but previous work found binary event meta-analytic results can be fragile, affected by only a small number of patients in specific trials. Meta-analyses can also miss literature, and a method for estimating how much additional unseen data would flip results would be a useful tool. This works establishes a complementary and generalisable definition of meta-analytic fragility, based on Ellipse of Insignificance (EOI) and Region of Attainable Redaction (ROAR) methods originally developed for dichotomous outcome trials. This method does not require trial-specific alterations to estimate fragility and yields a general method to estimate robustness of a meta-analysis to data redaction or addition of hypothetical trial outcomes. This method is applied to 3 meta-analyses with conflicting findings on the association of vitamin D supplementation and cancer mortality. A full meta-analysis of all trials cited in the 3 meta-analyses yielded no association between vitamin D supplementation and cancer mortality. Using the method outlined here, it was determined that meta-analytic fragility was high in all cases, with recoding of just 5 patients in the full cohort of 133,262 patients was enough to cross the significance threshold. Small amounts of redacted or non-included data also had substantial impact on each meta-analysis, with addition of just 3 hypothetical patients to an ostensibly significant meta-analyses (N = 38,538) enough to yield a null result. This method for analytical fragility is complementary to previous investigations that suggested meta-analyses are frequently fragile. It further shows that merely increasing the sample size is not an assurance against fragility. Caution should be advised when interpreting the results of meta-analyses and conflicting results may stem from inherent fragility and should be carefully employed.

Introduction

Meta-analyses are critical for synthesising medical evidence and providing robust estimates of treatment effects (Murad et al., 2014; Dechartres et al., 2013; Nordmann et al., 2012). However, there are concerns that meta-analyses are frequently improperly conducted, and in recent years there has been methodological concerns over the rigour of an increasing number of these publications (Lawrence et al., 2021; Siemens et al., 2021; Hameed et al., 2020), particularly when constituent Randomized Controlled Trials (RCTs) have data integrity issues which meta-analysis cannot solve (Wilkinson et al., 2024, 2025). There is an additional consideration that deserves mention particularly in the context of dichotomous outcome or binary outcome trials, the issue of fragility. In RCTs, many trial results are fragile, with recoding (the flipping of events to non-events or vice versa, in either or both arms) of a small number of events to non-events or vice versa in either the experimental or control arm often sufficient to convert a seemingly statistically significant result to a null, and vice versa (Grimes, 2022; Tignanelli and Napolitano, 2019; Baer et al., 2021a, Baer et al., 2021b).

The issue of fragility has generated much recent discussion in RCTs, with numerous studies on various types of RCTs demonstrating that many binary results are not robust, including in oncology trials (Del Paggio and Tannock, 2019), cardiovascular trials (Khan et al., 2019), ophthalmology trials (Shen et al., 2018), HIV research (Wayant et al., 2019), and gynecologic surgery (Pascoal et al., 2022). FI has not however received much consideration for meta-analyses, which are often thought of as more robust than RCTs. This however may not be the case. Previously, Atal et al (Atal et al., 2019) suggested meta-analyses may be more fragile than might be presumed in binary outcome cases. In that work, they established an algorithm to find the minimum re-coding of patients between studies that would engineer a seemingly significant result from a null one or vice versa. Applying this to a large sample of meta-analyses, Atal et al found that many were inherently fragile. More recently, Xing et al (Xing et al., 2025) noted that fragility in meta-analyses was an understudied problem, and employed Atal's method on 3758 meta-analyses to determine the mean fragility index (FI) was 5 (IQR: 2–11), with 15% of those studies having an FI of 1.

Atal's method is highly useful, but one possible objection is that it has the downside of non-generalisability, as it finds very specific combinations of trials and patients that would have to be re-coded (events classified as non-events and vice-versa) for results to become insignificant. For example, an Atal meta-analytic fragility of 4 pertains to a specific and often unique circumstance when 4 patients could be recoded from a specific study or combinations thereof to change outputs, but this does not generalise to any 4 patients in that meta-analysis. This makes this definition of meta-analytic fragility useful but not general, and perhaps less intuitive to interpret than a typical RCT fragility metric. In this work, we establish a generalizable meta-analytic fragility metric, based upon Ellipse of Insignificance (EOI) analysis for dichotomous outcome trials (Grimes, 2022). This method creates a pool of events and non-events in both arms, adjusted for weighing, and answers the general question of how many patients could be recoded in a meta-analyses for results to flip.

Ellipse of Insignificance and Region of Attainable Redaction

EOI is a refined analytical fragility metric that simultaneously finds the degree of fragility in both control and experimental arms of dichotomous outcome trials. Consider any 2×2 chi squared test with a events and b non-events in the experimental arm and c events and d non-events in the control arm at significance level α. EOI theory states there is an ellipse defined by these parameters, which can be mapped on a plane with axes corresponding to the experimental control arm. It further shows that any point inside that ellipse will be non-significant at α, and significant when outside the ellipse. For any points outside the ellipse (significant results) then EOI analysis analytically finds and resolves the minimum displacement (Fewest Experimental/Control Knowingly Uncoded Participants (FECKUP) vector) from the ellipse, and hence true fragility of any given dichotomous outcome experiment. Conversely, if a result is inside the ellipse (non-significant), EOI will also find the minimum recoding required to reject the null. EOI is flexible,analytical, and handles samples of all size rapidly and considers changes in the control and experimental arm simultaneously, rendering it a powerful tol for robustness testing.

Intimately related to this is Region of Attainable Redaction (ROAR) analysis (Grimes, 2024b), a methodology for ascertaining potential influence of redacted data on reported results. There are many reasons why relevant data might be missing from an analysis, from missed studies to inappropriate cut-offs and even deliberate cherry-picking. This is highly relevant to meta-analysis, and it would be useful to have a method to quantify the potential impact of redaction on reported results and gives an additional estimate of how robust reported results may be when new data is accrued. ROAR is based on similar principles to EOI, but the bounding shape produced by the theory is no longer an ellipse, but a more complex shape. Even so, like EOI, ROAR analysis analytically and rapidly measures the displacement from the shape boundary and thus the fragility of a result to hypothetical redaction of data as well as standard re-coding in traditional fragility analysis

Both EOI and ROAR are powerful tools for 2×2 dichotomous outcome trials, and their geometric and mathematical basis has been described preciously, as well as implemented in detecting potentially spurious results recently (Grimes and Heathers, 2025). Accordingly in this work, we will extend these methodologies to handle meta-analysis, contrast them to Atal et al's metric, and apply them to conflicting Vitamin D trials to illustrate their epidemiological utility. This package of meta-analytic fragility tools we name *EOIMETA* for brevity.

Methods

Derivation of Meta-analytic fragility and redaction analysis (EOIMETA)

For both EOI and ROAR, full mathematical details have been previously published (Grimes, 2022, 2024b) and an online implementation is available at *https://drg85.shinyapps.io/EOIROAR/* for any dichotomous outcome trial. Extending this to the meta-analytic situation, consider i studies, each with an experimental group with a_i events and b_i non-events, and control group with c_i events and d_i non-events per study. If we consider for any given meta-analysis the crude unadjusted pooled relative ratio (RR_p) and a Cochran–Mantel–Haenszel risk ratio (RR_{CMH}), we can define confounding effects between studies by $|1 - \frac{RR_{CMH}}{RR_p}|$. When this is small (< 10%), confounding is minimal and EOI analysis could be deployed on pooled meta-analysis results, by treating the entire meta-analysis as a single pooled sample. But as meta-analysis weigh studies by variance, simple pooling would not inherently account for study heterogeneity. Accordingly, we adjust using a generic inverse-variance weighed average model. For each included study, we compute relative risk and standard error as

$$\ln(RR_i) = \ln\left(\frac{a_i(c_i + d_i)}{c_i(a_i + b_i)}\right) \tag{1}$$

$$SE(\ln(RR_i)) = \sqrt{\frac{1}{a_i} + \frac{1}{a_i + b_i} + \frac{1}{c_i} + \frac{1}{c_i + d_i}} \tag{2}$$

where a continuity correction of 0.5 may be applied for any zero values. The weighted average of each trial is $w_i = SE(\ln(RR_i))^{-1}$ and total pooled risk ratio is given by

$$RR_{PG} = \exp\left(\frac{\Sigma w_i \ln(RR_i)}{\Sigma w_i}\right). \tag{3}$$

Once the pooled risk ratio is known, we can scale the event and non-event rate in the experimental group to match the adjusted relative risk while keeping the total number in the experimental arm constant. Letting $a_p = \Sigma a_i, b_p = \Sigma b_i, c_p = \Sigma c_i$ and $d_p = \Sigma d_i$, we allow $a_P^* = a + x$ and

$b_P^* = b - x$ where x may be positive or negative to adjust the experimental group to the derived risk ratio. We can solve for this value to obtain

$$x = \frac{c_P(a_P + b_P)RR_{PG}}{c_P + d_P} - a_P. \tag{4}$$

For consistency, the values of a_P^* and b_P^* are rounded to integers. This modification accounts for study level heterogeneity, and a standard EOI analysis can then be applied to the vector (a_P^*, b_P^*, c_P, d_P). In addition, we can also employ ROAR analysis to the same vector, yielding the raw number of patients in either or both arm who could be added a given direction to change the result, and exact combination of control and experimental group redactions required to change the result from a significant finding to a null one. Caveats for implementation and interpretation are outlined in the discussion section.

Application to vitamin D studies

Results of vitamin D supplementation and cancer mortality have been markedly inconsistent. In recent years, several meta-analyses of published trials yielded markedly conflicting results. A review and meta-analysis by Zhang et al (Zhang et al., 2019) and correction (Zhang et al., 2020) of 5 RCTs involving 38,538 patients found a significant association between vitamin D supplementation and reduced cancer mortality, reporting a 15% decrease in the risk of cancer death with vitamin D supplementation. Similarly, a meta-analysis by Gou et al (Guo et al., 2023) of 11 RCTs involving 111,952 patients found a significant reduction in relative risk of cancer death with supplementation of 12%. By contrast, a subsequent study of 6 RCTS involving 61,223 patients (Zhang et al., 2022) found no significant reduction in cancer death risk with vitamin D supplementation. All 3 meta-analyses involved combinations drawn from just 12 trials.

To investigate this apparent case of meta-analytic fragility, we extracted data from all 12 studies included in the 3 meta-analyses. A small inconsistency between all 3 meta-analysis was found in relation to one RCT (Scragg et al., 2018); In Zhang et al, the rate of cancer deaths in the supplementation and control group were reported as 44/2558 and 45/2550 respectively, versus being reported as 28/2558 versus 30/2558 in Gou et al and 30/2550 versus 30/2550 in Zheng et al. Disagreement between these figures appears to stem from cancers detected in both groups prior to the randomisation process. Accordingly, the adjusted ITT figure used in this work is 30/2544 versus 30/2535 respectively for Scragg et al's work. A further nuance is that for another small RCT (Ammann et al., 2017), direct data was not given on the number of cases. This however can be estimated from minimal assumption through the hazard ratio. For all included studies, relative risk and 95% confidence intervals were calculated explicitly except for Lehouck et al (Lehouck et al., 2012) where zero recorded deaths in the supplementation sample required confidence intervals to be estimated by bootstrap methods.Initially, we ran fragility analysis on all of these meta-analysis, with the new metric outlined here as well as Atal et al's algorithm. After this, we combined data and performed a meta-analysis on all 12 studies to ascertain potential effects of vitamin D and cancer mortality. A full fragility and robustness analysis was then performed on this, and results discussed to elucidate findings.

Meta-analytic fragility package

We developed a custom R package (Grimes, 2025) (eoirroar, available at *https://github.com/drg85/EOIMETA* and on Zenodo) to perform the fragility and redaction analysis described in the methods, as well as Atal et al's study specific alogorithm method for comparison.

Results

Meta-analytic fragility estimation of Vitamin D Cancer Mortality studies

For all conflicting meta-analyses, the unadjusted crude pooled risk ratio and CMH risk ratio were calculated prior to the analysis being conducted. For the three meta-analyses, this ranged from 0.08% – 1.1% confounding, corresponding to negligible difference. Table 1 gives the results for EOI and ROAR meta-analysis as well as the Atal et al method. Figure 2 depicts the EOI analysis results for meta-analysis by Zhang et al (including 5 RCTs, 38,538 patients) and Gou et al (11 RCTs, 111,952 patients).

Figure 1 gives the EOI meta-analytic fragility vectors for both positive meta-analyses, showing the degree of recoding in both experimental and control groups required to flip results. The negative values on both axes arise because the relative risk of the experimental group (vitamin D supplementation) is ostensibly lower than the placebo group, so in the convention of EOI analysis, negative values imply additional of events to the experimental group and/or subtraction of events from the control. In the case of Zhang et al, it would take recoding of just 4 events / non-events in either experimental or control arms, or 4 recodings total in combination (< 0.01% of the total sample) to lose significance, even less than the fragility estimated by Atal et al's algorithm. In the case of Guo et al, recoding of 38 events / non-events (< 0.038% of sample) achieves the same result, whereas there is a even more extreme specific fragility detected by Atal et al's algorithm.

The redaction analysis is also informative, because even if there were no coding or data errors in any study or their synthesis, the addition of small numbers of patients can profoundly change the result. In the case of Zhang et al, adding just 3 patients to the cohort would be enough to lose significance, while in Guo et al it is 37. Despite Zheng et al 2022 having substantially fewer patients than Guo et al 2022, it appears more robust to fragility by all metrics. This in turn leads to an important observation – the fact that a meta-analysis has more patients and studies does not inherently make it more robust or less fragile, as discussed in the next section.

Meta-analysis of all studies

Figure 2 depicts the meta-analysis forest plot of all 12 RCTs of 133,262 patients using a random effects model, while table 2 gives summary meta-analytic fragility statistics. Table 2 gives the statistics for when all 12 studies are combined, finding that both EOI fragility and Atal et al's fragility actually increase despite the increase in the number of subjects. Even with a large sample, the recoding of just 5 non-events to events, in this case in the placebo arm (additional deaths in the non-vitamin D group), would alter the results to cross the threshold of significance (p = 0.048) and likely change inferences. Table 3 gives both fragility estimates for each included study on an individual basis for comparison. Note that traditional the fragility index is not defined for non-significant results, which EOI fragility is sufficently flexible to find the counts needed to flip significance, including recoding required to move from non-significant to significant.

Discussion

The meta-analytic fragility tool presented here, EOIMETA, is generalisable and flexible, complementing Atal et al's study and patient specific method. The nuance between both approaches is that in Atal et al's algorithm identifies a specific combination of patients from specific studies that may be moved to alter results, whereas the current work establishes a method for pooling meta-analysis on ascertaining their group fragility and the potential impact of missing data. In this respect, they answer complementary and subtlety different questions specific to meta-analyses.

In RCT literature, a frequent objection to the use of fragility analysis methods is that the mere existence of a small FI might be an artefact of trial design. For clinical trials especially, a well-designed trial should be structured as to minimize patient exposure to unknown harms, striving to

Table 1. **Application of meta-analytic fragility / redaction sensitivity to conflicting meta-analyses**

Meta-analysis	Total patients	Risk Ratio (95% CI)	I^2 (95% CI)	P-value	EOI fragility	Redaction fragility	Atal fragility
Zhang et al 2019	38,538	0.85* (0.74 - 0.97)	0% (0% – 79.2%)	0.0349*	4	3	5
Guo et al 2022	111,952	0.87* (0.80 - 0.95)	0% (0% – 60.2%)	0.003*	38	37	33
Zhang et al 2022	61,223	0.96 (0.80 - 1.16)	53.5% (0% – 81.4%)	0.6624	51	96	50

Figure 1. **Meta-analytic EOI fragility analysis results for (a) Zhang et al and (b) Guo et al.**

See text for details.

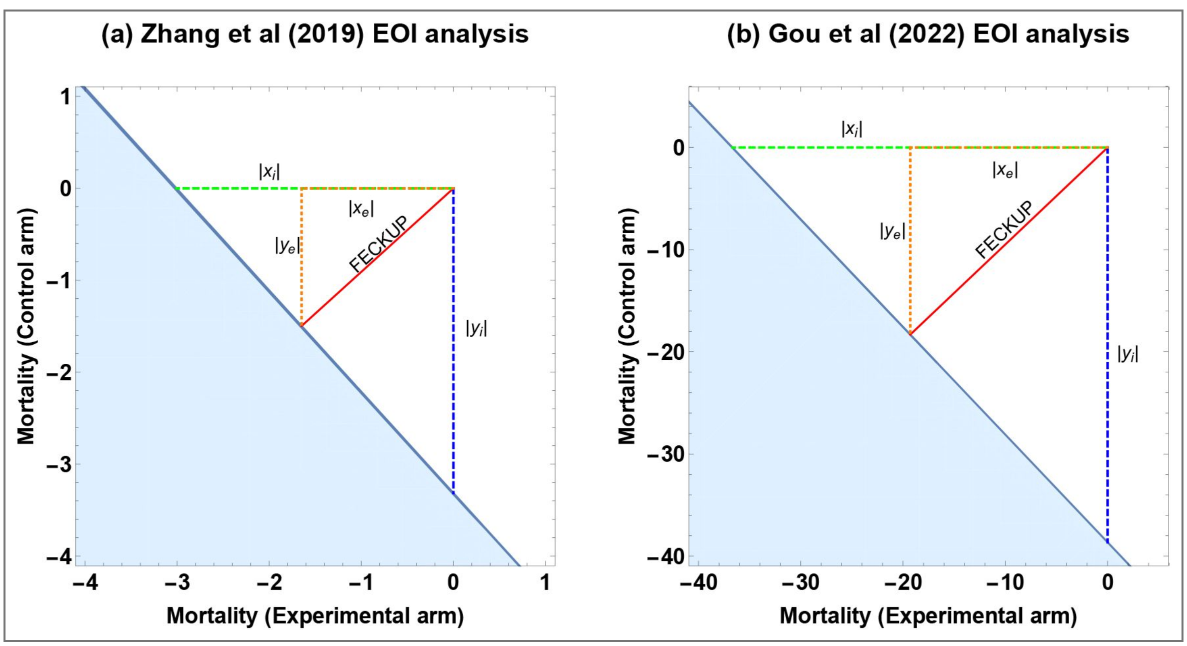


Figure 2. **Meta-analysis of all 12 RCTS including study details and relative risk of vitamin D supplementation on cancer mortality, shown with 95% confidence intervals.**

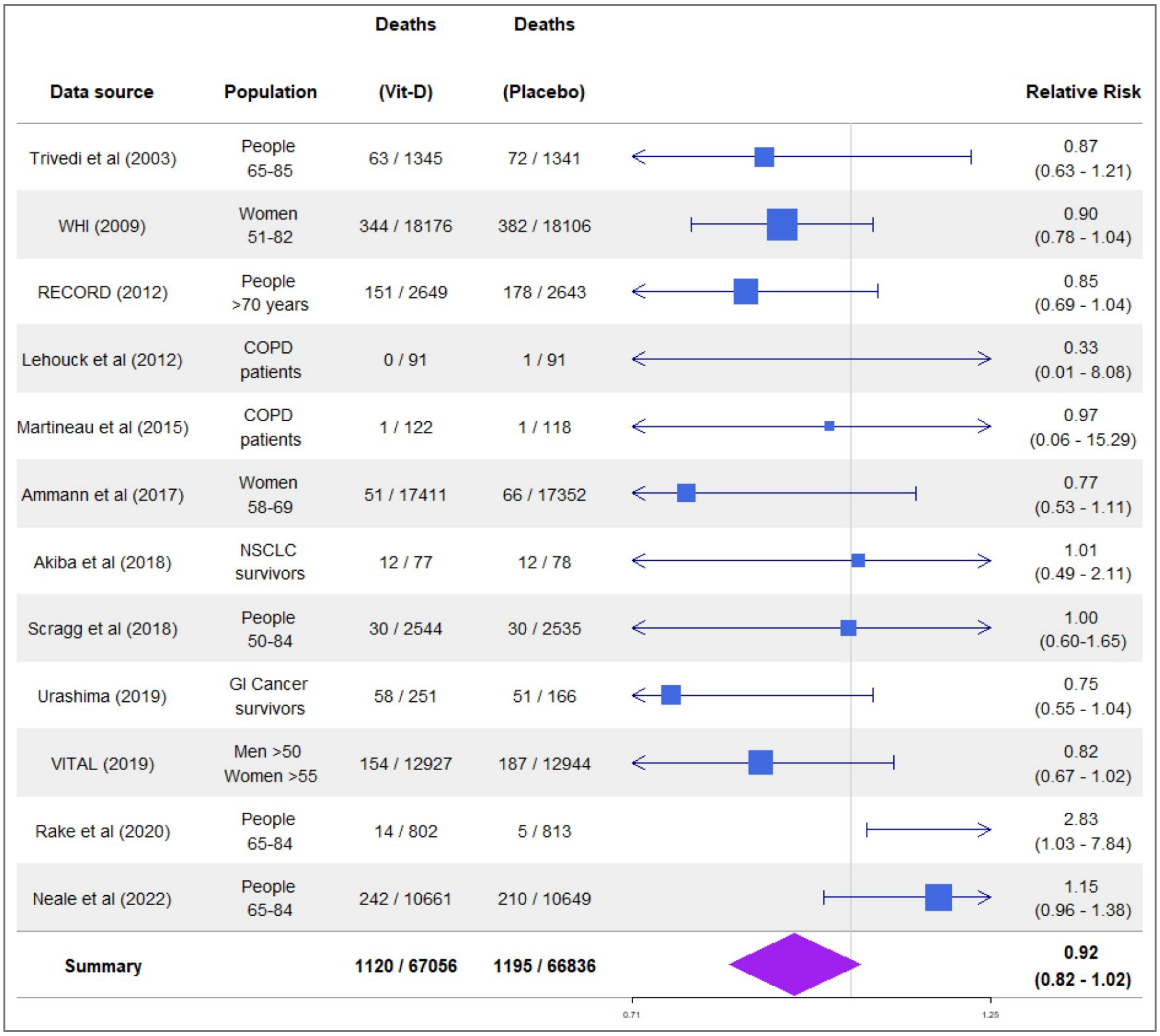


Table 2. **Meta-analytic fragility of all 12 studies**

Total patients	Risk Ratio (95% CI)	I^2 (95% CI)	P-value	EOI fragility	Redaction fragility	Atal fragility
133,262	0.92 (0.83 - 1.02)	24.9% (0% – 61.6%)	0.061	5 (0.00375%)	192 (0.14%)	9 (0.00675%)

Study	Deaths (Vitamin D)	Deaths (Placebo)	Fragility Index†	EOI fragility
Trivedi et al (2003)	63 / 1345	72 / 1341	NA	13
WHI (2009)	344 / 18176	382 / 18106	NA	13
RECORD (2012)	151 / 2649	178 / 2643	NA	8
Lehouck et al (2012)	0 / 91	1 / 91	NA	NA$^{\lambda}$
Martineau et al (2015)	1 / 122	1 / 118	NA	NA$^{\lambda}$
Ammann et al (2017)	51 / 17411	66 / 17352	NA	6
Akida et al (2018)	12 / 77	12 / 78	NA	9
Scragg et al (2019)	30 / 2544	30 / 2535	NA	15
Urashima (2019)	58 / 251	51 / 166	NA	2
VITAL (2019)	154 / 12927	187 / 12944	NA	12
Rake et al (2020)	14 / 802	5 / 813	1	1
Neale et al (2022)	242 / 10661	210 / 10649	NA	10

† Traditional FI not applicable when results are non-significant.

λ EOI Fragility ambiguous (multipe solutions) with zero or unity event counts

Table 3. **Fragilty comparison across all 12 studies (study-level metrics)**

ensure that just enough participate to allow the detection of clinically relevant effects. The argument that RCTs at least might be fragile ‘by design’ is however convincingly countered by other authors, who find no evidence of p-value distributions clustering around significance thresholds after a sample size calculation and find additional that fragility in well-designed studies is not always low. More recent robust methods for fragility analysis like EOI and for redaction like ROAR also have application far beyond ostensibly fragile-by-design RCTs, but for cohort studies, preclinical work, and even ecological studies. The fragile-by-design argument is also inherently weaker in the context of meta-analysis, where the marshalling of many studies to increase effect precision measurements should strengthen the evidence but results here and previously suggest that even meta-analysis of large groups of patients can be profoundly fragile.

There are a few subtle issues and limitations to consider additionally; Atal et al’s meta-analytic fragility metric in principle identifies the precise combination of inter-study and patient recoding required to flip a result, a “worse-case” scenario, identifying the most fragile point in a meta-analyses. This usually results in the Atal et al fragility metric being typically smaller than the complement EOI fragility metric outlined here, but this is not always the case. Because Atal et al’s method is a greedy algorithm, it is time-consuming to run on large collections of studies and can sometimes miss optimal and smaller solutions. Secondly, the adjustment of the events and non-events in the experimental group due to weighed inverse variance methods in the EOI fragility method subtly alters group composition, resulting in a smaller EOI fragility than a crude pooled estimate. What is evident from the examples shown in this work is that even in large cohorts with relatively low heterogeneity, the impact of recoding only miniscule numbers of patients can drastically alter interpretation; in the 12-study meta-analysis, recoding a mere 5 patients in 133,262 was sufficient to alter the result from a null to a positive finding.

While the method outlined here is analytic and rapid, it has some limitations that must be considered. Like Atal’s method, it applies only to meta-analyses of trials with dichotomous outcomes. It should be noted there are also fragility indices designed for continuous data (Caldwell et al., 2021) but these are beyond the scope of the current work. While EOIMETA has a generic weighed inverse variance (WIV) correction by default, it may not be reliable in the presence of high between-study heterogeneity. Accordingly, careful application is required. At its most essential, this implementation reflects the meta-analytic relative risk by generating a synthetic pooled 2×2 table that modifies pooled events (and non-events) in the experimental arm so that trial results may be pooled and meta-analytic fragility estimated. One interesting feature of the current method is that neglecting WIV correction and simply pooling trials for basic EOI or ROAR analysis may be justifable in some instances, especially when $RR_p \approx RR_{CMH}$. The main effect of this simpliciation is that it produces much more fragile estimates than those presented here; for example, a simple pooling EOI analysis on Guo et al produces an EOI of 17 (as opposed to the 38 reported here) and a ROAR of just 16 (versus the 96 reported here). Thus the results reported here are thus conservative, though the precise interpretation of these variants in implementation is an open question.

The meta-analytic fragility metric is useful for modeling perturbations in results, such as the addition of patients (who were previously redacted) or the recoding of events and non-events. The advantage of this approach lies in its tractability, offering a simpler alternative to working with multiple studies individually and a deterministic and analytic resultant fragility metric. Because the resultant metric pertains to the effective pool of patients in the meta-analysis, it does not identify individual studies that may unduly influence results. For this purpose, leave-one-out analysis (Willis and Riley, 2017) or extensions (Meng et al., 2024) are more suitable tools, complementing the method described in this work.

The use of vitamin D meta-analyses in this work was chosen as illustrative rather than specific, but it is worth noting that there are methodological concerns with much Vitamin D research (Grimes et al., 2024). The three studies cited in this work report relatively low heterogeneity in their meta-analysis in both effect sizes and I^2 values, but it is worth noting that the included studies addressed very different populations, including patients with Chronic Obstructive Pulmonary Disease, Non-small cell lung cancer survivors, women only cohorts, older adults, and gastrological cancer survivors. These groups have presumably different risk factors for cancer deaths, and why the authors of these studies combined the cohorts with fundamentally different clinical contexts is unclear. Why the heterogeneity appeared so relatively low in different groups is also a curious feature. This goes beyond the scope of the current work, but serves as an example of the reality that meta-analysis is only as strong as its underlying data and methodological rigor in comparing like-with-like, and the conclusions drawn from them must always be seen in context.

While meta-analysis is a powerful method for refining effect size estimates, it cannot overcome poorly conducted research or bad data (Jané et al., 2025), nor it is intrinsically robust. These limitations should be kept in mind when considering meta-analytic results, and scientists should consider them doubly when opting to perform a meta-analysis. Mass produced and unreliable meta-analyses are a recognised and growing problem (Ioannidis, 2016), and we need be mindful not to add to increasingly issue of research waste and unreliable research (Glasziou and Chalmers, 2018; Grimes, 2024a).

Data availability

All relevant code for this undertaking including the study data from the included trials is available at the linked Github repository and the Zenodo DOI, as well as additional EOI and ROAR plots stemming from this work. This article was previously uploaded as a preprint to Medrxiv (doi: 10.1101/2025.08.15.25333793).

Additional information

Author ORCID iDs

David Robert Grimes: https://orcid.org/0000-0003-3140-3278

References

Ammann EM, Drake MT, Haraldsson B, Wallace RB, Johnson KC, Desai P, Lin EM, Link BK (2017) Incidence of hematologic malignancy and cause-specific mortality in the Women's Health Initiative randomized controlled trial of calcium and vitamin D supplementation. *Cancer* **123**:4168-4177

Atal I, Porcher R, Boutron I, Ravaud P. (2019) The statistical significance of meta-analyses is frequently fragile: definition of a fragility index for meta-analyses. *Journal of Clinical Epidemiology* **111**:32-40

Baer BR, Gaudino M, Charlson M, Fremes SE, Wells MT (2021a) Fragility indices for only sufficiently likely modifications. *Proceedings of the National Academy of Sciences* **118**:e2105254118

Baer BR, Gaudino M, Fremes SE, Charlson M, Wells MT (2021b) The fragility index can be used for sample size calculations in clinical trials. *Journal of Clinical Epidemiology* **139**:199-209

Caldwell JME, Youssefzadeh K, Limpisvasti O. (2021) A method for calculating the fragility index of continuous outcomes. *Journal of Clinical Epidemiology* **136**:20-25

Dechartres A, Trinquart L, Boutron I, Ravaud P. (2013) Influence of trial sample size on treatment effect estimates: meta-epidemiological study. *Bmj* **346**

Del Paggio JC, Tannock IF (2019) The fragility of phase 3 trials supporting FDA-approved anticancer medicines: a retrospective analysis. *The Lancet Oncology* **20**:1065-1069

Glasziou P, Chalmers I. (2018) Research waste is still a scandal—an essay by Paul Glasziou and Iain Chalmers. *Bmj* **363**

Grimes DR (2022) The ellipse of insignificance, a refined fragility index for ascertaining robustness of results in dichotomous outcome trials. *eLife* **11**:e79573 https://doi.org/10.7554/eLife.79573

Grimes DR (2024a) Is biomedical research self-correcting? Modelling insights on the persistence of spurious science. *Royal Society Open Science* **11**:231056

Grimes DR (2024b) Region of Attainable Redaction, an extension of Ellipse of Insignificance analysis for gauging impacts of data redaction in dichotomous outcome trials. *eLife* **13**:e93050 https://doi.org/10.7554/eLife.93050

Grimes DR (2025) EOIMETA: EOIMETA V1 R code (eoimeta). Zenodo. https://doi.org/10.5281/zenodo.15878923

Grimes DR, Heathers J. (2025) "Cake causes herpes?"-promiscuous dichotomisation induces false positives. *BMC Medical Research Methodology* **25**:1-10

Grimes DR, Kimlin M, Glasziou P. (2024) Arbitrary vitamin D deficiency thresholds yield unreliable and potentially spurious results–a review and investigation. *Open Science Preprints* https://doi.org/10.31219/osf.io/su6cx

Guo Z, Huang M, Fan D, Hong Y, Zhao M, Ding R, Cheng Y, Duan S. (2023) Association between vitamin D supplementation and cancer incidence and mortality: A trial sequential meta-analysis of randomized controlled trials. *Critical Reviews in Food Science and Nutrition* **63**:8428-8442

Hameed I, Demetres M, Tam DY, Rahouma M, Khan FM, Wright DN, Mages K, DeRosa AP, Baltich Nelson B, Pain K, *et al.* (2020) An assessment of the quality of current clinical meta-analyses. *BMC Medical Research Methodology* **20**:105

Ioannidis JP (2016) The mass production of redundant, misleading, and conflicted systematic reviews and meta-analyses. *The Milbank Quarterly* **94**:485-514

Jané MB, Heathers J, Grimes DR (2025) Major Flaws in Taylor et al.'s (2025) Meta-analysis on Fluoride Exposure and Children's IQ Scores. *Open Science Preprints* https://doi.org/10.31219/osf.io/zhm54_v3

Khan MS, Ochani RK, Shaikh A, Usman MS, Yamani N, Khan SU, Murad MH, Mandrola J, Doukky R, Krasuski RA (2019) Fragility index in cardiovascular randomized controlled trials. *Circulation: Cardiovascular Quality and Outcomes* **12**:e005755

Lawrence JM, Meyerowitz-Katz G, Heathers JA, Brown NJ, Sheldrick KA (2021) The lesson of ivermectin: meta-analyses based on summary data alone are inherently unreliable. *Nature Medicine* **27**:1853-1854

Lehouck A, Mathieu C, Carremans C, Baeke F, Verhaegen J, Van Eldere J, Decallonne B, Bouillon R, Decramer M, Janssens W. (2012) High doses of vitamin D to reduce exacerbations in chronic obstructive pulmonary disease: a randomized trial. *Annals of internal medicine* **156**:105-114

Meng Z, Wang J, Lin L, Wu C. (2024) Sensitivity analysis with iterative outlier detection for systematic reviews and meta-analyses. *Statistics in medicine* **43**:1549-1563

Murad MH, Montori VM, Ioannidis JP, Jaeschke R, Devereaux P, Prasad K, Neumann I, Carrasco-Labra A, Agoritsas T, Hatala R, *et al.* (2014) How to read a systematic review and meta-analysis and apply the results to patient care: users' guides to the medical literature. *Jama* **312**:171-179

Nordmann AJ, Kasenda B, Briel M. (2012) Meta-analyses: what they can and cannot do. *Swiss medical weekly* **142**:w13518-w13518

Pascoal E, Liu M, Lin L, Luketic L. (2022) The fragility of statistically significant results in gynaecologic surgery: a systematic review. *Journal of Obstetrics and Gynaecology Canada* **44**:508-514

Scragg R, Khaw KT, Toop L, Sluyter J, Lawes CM, Waayer D, Giovannucci E, Camargo CA (2018) Monthly high-dose vitamin D supplementation and cancer risk: a post hoc analysis of the vitamin D assessment randomized clinical trial. *JAMA oncology* **4**:e182178-e182178

Shen C, Shamsudeen I, Farrokhyar F, Sabri K. (2018) Fragility of results in ophthalmology randomized controlled trials: a systematic review. *Ophthalmology* **125**:642-648

Siemens W, Schwarzer G, Rohe MS, Buroh S, Meerpohl JJ, Becker G. (2021) Methodological quality was critically low in 9/10 systematic reviews in advanced cancer patients—A methodological study. *Journal of Clinical Epidemiology* **136**:84-95

Tignanelli CJ, Napolitano LM (2019) The fragility index in randomized clinical trials as a means of optimizing patient care. *JAMA surgery* **154**:74-79

Wayant C, Meyer C, Gupton R, Som M, Baker D, Vassar M. (2019) The fragility index in a cohort of HIV/AIDS randomized controlled trials. *Journal of General Internal Medicine* **34**:1236-1243

Wilkinson J, Heal C, Antoniou GA, Flemyng E, Alfirevic Z, Avenell A, Barbour G, Brown NJ, Carlisle J, Clarke M, *et al.* (2024) Protocol for the development of a tool (INSPECT-SR) to identify problematic randomised controlled trials in systematic reviews of health interventions. *BMJ open* **14**:e084164

Wilkinson J, Heal C, Antoniou GA, Flemyng E, Ahnström L, Alteri A, Avenell A, Barker TH, Borg DN, Brown NJ, *et al.* (2025) Assessing the feasibility and impact of clinical trial trustworthiness checks via an application to Cochrane Reviews: Stage 2 of the INSPECT-SR project. *Journal of Clinical Epidemiology* 111824

Willis BH, Riley RD (2017) Measuring the statistical validity of summary meta-analysis and meta-regression results for use in clinical practice. *Statistics in medicine* **36**:3283-3301

Xing A, Xing X, Murad MH, Lin L. (2025) Evaluating the properties of the fragility index of meta-analyses. *BMC Medical Research Methodology* **25**:212

Zhang R, Zhang Y, Liu Z, Pei Y, Xu P, Chong W, Hai Y, He L, He Y, Yu J, *et al.* (2022) Association between vitamin D supplementation and cancer mortality: a systematic review and meta-analysis. *Cancers* **14**:3717

Zhang Y, Fang F, Tang J, Jia L, Feng Y, Xu P, Faramand A. (2019) Association between vitamin D supplementation and mortality: systematic review and meta-analysis. *bmj* **366**

Zhang Y, Fang F, Tang J, Jia L, Feng Y, Xu P, Faramand A. (2020) Correction: Association between vitamin D supplementation and mortality: systematic review and meta-analysis. *bmj* **370**

Peer reviews

Reviewer #1 (Public review):

Summary:

This manuscript addresses an important methodological issue-the fragility of meta-analytic findings-by extending fragility concepts beyond trial-level analysis. The proposed EOIMETA framework provides a generalizable and analytically tractable approach that complements existing methods such as the traditional Fragility Index and Atal et al.'s algorithm. The findings are significant in showing that even large meta-analyses can be highly fragile, with results overturned by very small numbers of event recodings or additions. The evidence is clearly presented, supported by applications to vitamin D supplementation trials, and contributes meaningfully to ongoing debates about the robustness of meta-analytic evidence. Overall, the strength of evidence is moderate to strong.

Strengths:

(1) The manuscript tackles a highly relevant methodological question on the robustness of meta-analytic evidence.

(2) EOIMETA represents an innovative extension of fragility concepts from single trials to meta-analyses.

(3) The applications are clearly presented and highlight the potential importance of fragility considerations for evidence synthesis.

https://doi.org/10.7554/eLife.108693.2.sa2

Reviewer #3 (Public review):

Summary and strengths:

In this manuscript, Grimes presents an extension of Ellipse of Insignificant (EOI) and Region of Attainable Redaction (ROAR) metrics to meta-analysis setting as metrics for fragility and robustness evaluation of meta-analysis. The author applies these metrics to three meta-analyses of Vitamin D and cancer mortality, finding substantial fragility in their conclusions. Overall, I think extension/adaption is a conceptually valuable addition to meta-analysis evaluation, and the manuscript is generally well-written.

Specific comments:

(1) The manuscript would benefit from a clearer explanation of in what sense EOIMETA is generalizable. The author mentions this several times, but without a clear explanation of what they mean here.

(2) The authors mentioned the proposed tools assume low between-study heterogeneity. Could the author illustrate mathematically in the paper how the between-study heterogeneity would influence the proposed measures? Moreover, the between-study heterogeneity is high in Zhang et al's 2022 study. It would be a good place to comment on the influence of such high heterogeneity on the results, and specifying a practical heterogeneity cutoff would better guide future users.

(3) I think clarifying the concepts of "small effect", "fragile result", and "unreliable result" would be helpful for preventing misinterpretation by future users. I am concerned that the audience may be confusing these concepts. A small effect may be related to a fragile meta-analysis result. A fragile meta-analysis doesn't necessarily mean wrong/untrustworthy results. A fragile but precise estimate can still reflect a true effect, but whether that size of true effect is clinically meaningful is another question. Clarifying the effect magnitude, fragility, and reliability in the discussion would be helpful.

Comments on revisions:

I am unable to find the author's responses to my previous round comments (Reviewer #3) in the revision package, though replies to the other reviewers are present. I will provide my updated feedback once these responses are available for review.

https://doi.org/10.7554/eLife.108693.2.sa1

Author response:

The following is the authors' response to the original reviews

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *This manuscript addresses an important methodological issue - the fragility of meta-analytic findings - by extending fragility concepts beyond trial-level analysis. The proposed EOIMETA framework provides a generalizable and analytically tractable approach that complements existing methods such as the traditional Fragility Index and Atal et al.'s algorithm. The findings are significant in showing that even large meta-analyses can be highly fragile, with results overturned by very small numbers of event recodings or additions. The evidence is clearly presented, supported by applications to vitamin D supplementation trials, and contributes meaningfully to ongoing debates about the robustness of meta-analytic evidence. Overall, the strength of evidence is moderate to strong, though some clarifications would further enhance interpretability.*

Strengths:

(1) The manuscript tackles a highly relevant methodological question on the robustness of meta-analytic evidence.

(2) EOIMETA represents an innovative extension of fragility concepts from single trials to meta-analyses.

(3) The applications are clearly presented and highlight the potential importance of fragility considerations for evidence synthesis.

Weaknesses:

(1) The rationale and mathematical details behind the proposed EOI and ROAR methods are insufficiently explained. Readers are asked to rely on external sources (Grimes, 2022; 2024b) without adequate exposition here. At a minimum, the definitions, intuition, and key formulas should be summarized in the manuscript to ensure comprehensibility.

(2) EOIMETA is described as being applicable when heterogeneity is low, but guidance is missing on how to interpret results when heterogeneity is high (e.g., large I^2*). Clarification in the Results/Discussion is needed, and ideally, a simulation or illustrative example could be added.*

(3) The manuscript would benefit from side-by-side comparisons between the traditional FI at the trial level and EOIMETA at the meta-analytic level. This would contextualize the proposed approach and underscore the added value of EOIMETA.

(4) Scope of FI: The statement that FI applies only to binary outcomes is inaccurate. While originally developed for dichotomous endpoints, extensions exist (e.g., Continuous Fragility Index, CFI). The manuscript should clarify that EOIMETA focuses on binary outcomes, but FI, as a concept, has been generalized.

Reviewer #2 (Public review):

Summary:

The study expands existing analytical tools originally developed for randomized controlled trials with dichotomous outcomes to assess the potential impact of missing data, adapting them for meta-analytical contexts. These tools evaluate how missing data may influence meta-analyses where p-value distributions cluster around significance thresholds, often leading to conflicting meta-analyses addressing the same research question. The approach quantifies the number of recodings (adding events to the experimental group and/or removing events from the control group) required for a meta-analysis to lose or gain statistical significance. The author developed an R package to perform fragility and redaction analyses and to compare these methods with a previously established approach by Atal et al. (2019), also integrated into the package. Overall, the study provides valuable insights by applying existing analytical tools from randomized controlled trials to meta-analytical contexts.

Strengths:

The author's results support his claims. Analyzing the fragility of a given meta-analysis could be a valuable approach for identifying early signs of fragility within a specific topic or body of evidence. If fragility is detected alongside results that hover around the significance threshold, adjusting the significance cutoff as a function of sample size should be considered before making any binary decision regarding statistical significance for that body of evidence. Although the primary goal of meta-analysis is *effect estimation, conclusions often still rely on threshold-based interpretations, which is understandable. In some of the examples presented by Atal et al. (2019), the event recoding required to shift a meta-analysis from significant to non-significant (or vice versa) produced only minimal changes in the effect size estimation. Therefore, in bodies of evidence where meta-analyses are fragile or where results cluster near the null, it may be appropriate to adjust the cutoff. Conducting such analyses-identifying fragility early and adapting thresholds accordingly-could help flag fragile bodies of evidence and prevent future conflicting meta-analyses on the same question, thereby reducing research waste and improving reproducibility.*

Weaknesses:

It would be valuable to include additional bodies of conflicting literature in which meta-analyses have demonstrated fragility. This would allow for a more thorough assessment of the consistency of these analytical tools, their differences, and whether this particular body of literature favored one methodology over another. The method proposed by Atal et al. was applied to numerous meta-analyses and demonstrated consistent performance. I believe there is room for improvement, as both the EOI and ROAR appear to be very promising tools for identifying fragility in meta-analytical contexts.

I believe the manuscript should be improved in terms of reporting, with clearer statements of the study's and methods' limitations, and by incorporating additional bodies of evidence to strengthen its claims.

Reviewer #3 (Public review):

Summary and strengths:

In this manuscript, Grimes presents an extension of the Ellipse of Insignificant (EOI) and Region of Attainable Redaction (ROAR) metrics to the meta-analysis setting as metrics for fragility and robustness evaluation of meta-analysis. The author applies these metrics to three meta-analyses of Vitamin D and cancer mortality, finding substantial fragility in their conclusions. Overall, I think extension/adaptation is a conceptually valuable addition to meta-analysis evaluation, and the manuscript is generally well-written.

Specific comments:

(1) The manuscript would benefit from a clearer explanation of in what sense EOIMETA is generalizable. The author mentions this several times, but without a clear explanation of what they mean here.

(2) The authors mentioned the proposed tools assume low between-study heterogeneity. Could the author illustrate mathematically in the paper how the between-study heterogeneity would influence the proposed measures? Moreover, the between-study heterogeneity is high in Zhang et al's 2022 study. It would be a good place to comment on the influence of such high heterogeneity on the results, and specifying a practical heterogeneity cutoff would better guide future users.

(3) I think clarifying the concepts of "small effect", "fragile result", and "unreliable result" would be helpful for preventing misinterpretation by future users. I am concerned that the audience may be confusing these concepts. A small effect may be related to a fragile meta-analysis result. A fragile meta-analysis doesn't necessarily mean wrong/untrustworthy results. A fragile but precise estimate can still reflect a true effect, but whether that size of true effect is clinically meaningful is another question. Clarifying the effect magnitude, fragility, and reliability in the discussion would be helpful.

I am very appreciative of the insightful comments you all shared, and in light of them have made several clarifications and revisions. Thank you again, I am grateful to have received such considered feedback and I hope I've addressed any outstanding issues. I have replied to each reviewer's recommendations in this document sequentially for ease of scanning, and am most grateful for the summary strengths and weaknesses, which I am also incorporated into these replies. Thank you again!

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *(1) The manuscript makes the important argument that many meta-analyses are inherently fragile, which aligns with prior work (e.g., PMID: 40999337). Please add the reference to the statements.*

Excellent point, thank you – I've expanded the discussion of fragility analysis, and its application to meta-analysis, including this reference.

> *(2) The rationale and mathematical underpinnings of the proposed EOI and ROAR methods are not sufficiently explained. While the authors cite Grimes (2022, 2024b), readers are expected to rely heavily on these external sources without adequate exposition in the current paper. This limits the ability to fully evaluate the reasonableness of the methods or to reproduce the approach. I strongly recommend expanding the description of EOI and ROAR within the manuscript.*

I agree fully – I was a little remiss in this scope, as I was worried about overwhelming the reader. However, I was too sparse with detail and have now extended the text this way to describe the methods intuitively as possible (see Discussion, subsection "Ellipse of Insignificance and Region of Attainable Redaction"

> *(3) In the Methods, the authors note that EOIMETA is applicable when between-study heterogeneity is low. However, the manuscript provides little guidance on how to interpret results when heterogeneity is high (e.g., larger I^2 values). I recommend clarifying this issue in the Results or Discussion sections, emphasizing the limitations of EOIMETA under high heterogeneity. Ideally, the authors could include either a small simulation study or an illustrative example to demonstrate the performance of the method in such settings.*

This is an excellent question, and I was remiss for not considering it better in the manuscript. Originally, the simple idea was to just pool the results for EOI, in which case heterogeneity would be an issue. But I then subsequently added weighed-inverse variance methods to account for situations with increased heterogeneity, so my initial comment was not strictly correct. I've changed the text in several places, notably in the methods and in the discussion (see reply point 5).

> *(4) While EOIMETA is introduced as a generalizable fragility metric for meta-analyses, the illustrative examples would benefit from clearer comparisons with the traditional Fragility Index (FI). Because FI is well established in the RCT literature and familiar to many readers, presenting side-by-side results (e.g., FI at the trial level versus EOIMETA at the meta-analytic level) would provide important context. Such comparisons would also highlight the added value of EOIMETA, underscoring that even when individual trials appear robust under FI, the pooled meta-analysis may remain fragile.*

This is an excellent idea! The new table is given below. Note that traditional FI are not defined for non-significant results, and EOI is ambiguous for counts <2.

> *(5) In the Discussion currently states that the Fragility Index (FI) applies only to binary outcomes. This is not entirely accurate. While the original FI was indeed developed for dichotomous endpoints, subsequent methodological work has extended the concept to other data types, including continuous outcomes (continuous fragility index, CFI). The manuscript should acknowledge this distinction: EOIMETA presently focuses on binary outcomes at the meta-analytic level, but FI more broadly is not restricted to binary data.*

> *Adding this clarification, with appropriate citations, would improve accuracy and place EOIMETA more clearly within the broader fragility literature.*

Thank you for this catch – clarified now in the discussion:

> ***Reviewer #2 (Recommendations for the authors):***
>
> *(1) Typos/inconsistencies/writing clarifications: All table and figure legends and titles are missing a period at the end of each sentence. In the sentence "to be estimated by bootstrap methods. Initially, we ran...", there should be a space between "methods" and "Initially" (line 113).*

Apologies, these are now remedied.

> *(2) In Table 2, the total number of patients in the meta-analysis of all 12 studies is reported as 133,262, whereas the text states 133,475 patients. Based on my calculations from Figure 2, the total appears to be 133,262. Could you please clarify this discrepancy?*

Certainly – your calculations are correct. The text figure was a typo based on a very early draft where the summation function was not correctly run, and doubled counted some cases. This was fixed for the figure but not the text. The text should now match, thank you for spotting this. There are some issues with figure 2, which I will address in next few points.

> *(3) Regarding this point, the meta-analysis by Zhang et al. (2019) shows some inconsistencies in the reported number of patients in the paper. According to the data provided on GitHub the total number of patients is 37671. However, Table 1 of the paper lists 38538 patients, and the main text states "5 RCTs involving 39168 patients." Similarly, for Guo et al. (2023), the main text reports that the meta-analysis included 11 RCTs with 112165 patients, whereas the table lists 111952, which appears consistent with the data available on GitHub. There is also a discrepancy in Zhang et al. (2022), which cites 61853 patients in the introduction but 61223 patients in Table 1. These inconsistencies should be clarified, as even small discrepancies in reported sample sizes can undermine the credibility of the analyses presented.*

Well-spotted – the incorrect figures are artefacts of an early draft with a double-counting summation function, and I should have spotted them and removed them prior to submission. To clarify, the correct figures from each study (which agree with github data) are given in the corrected table 1.

Thus, there are 38,538 subjects in the Zhang et al 2019 analysis, which matches the first sheet of the github listing. The confusion comes from sheet 2 which was included only with this, which breaks these events down into events / non-events (hence the total non-events being 37,671) but keeps the old labels. This is needlessly confusing, and accordingly I have re-uploaded the data with correct headers for sheet 2. This summation problem was also apparent in the total of figure 2, which has been replaced with a correct version now. Thank you for spotting this!

> *(4) In line 158, who does "He" refer to? Please clarify this in more detail.*

Apologies, this was a typo and should have read "the" – now corrected.

> *(5) The discrepant results of the RCT by Scragg et al. (2018) between the meta-analysis by Zhang et al. and that by Guo et al. could be presented in a table. This could be included as supplementary material or, preferably, in the main text (Results section).*

To avoid confusion, I will add a version of this to the github files for interested users to explore.

> *(6) In the legend of Figure 2, a period is missing at the end of the sentence. Additionally, although it is generally understood, it would be helpful to specify that the numbers in parentheses represent the confidence intervals. Please confirm whether these are 95%, 89%, or 99% confidence intervals.*

Apologies, these are 95% CIs. Clarified now in updated legends.

> *(7) The statement of "The more recent and robust methods for fragility analysis (EOI) and redaction (ROAR) have potential applications beyond fragile-by-design RCTs, extending to cohort studies, preclinical work, and even ecological studies, as stated by the author" in line 163. Could you please provide references supporting these claims? I believe the relevant references may be included in the EOI paper, but it would be helpful to cite them here as well.*

This has recently been used in new analysis now cited in the introduction with fuller description of method for context. Please see response to reviewer 1, points 2

> *(8) Since the study was previously published as a preprint (https://www.medrxiv.org/content/10.1101/2025.08.15.25333793v1.full-text), this should be mentioned in the manuscript.*

Added as a note now.

> *(9) It would also be valuable to include a figure illustrating ROAR for the same meta-analyses presented in Figure 1 for EOI, possibly as supplementary material.*

See reply to point 10.

> *(10) Finally, it would be interesting to provide plots of both EOI and ROAR for the meta-analyses of all 12 included studies. These graphs could be replicated using the code examples provided by the author in the original EOI and ROAR publications.*

These have now been added to the github repository as supplementary material.

> *(11a) Replications of EOI fragility: eoicfunc.R (github): - In the code provided on GitHub, an error occurred in the "EllipseFromEquation" function within eoifunc. This was due to the PlaneGeometry package not being available for the latest version of R. I attempted several installation methods (using devtools, remotes, and GitHub, as well as direct installation from a URL). However, after adjusting the code, I was able to run the analyses. For the full cohort, including all 12 studies using the EOI approach, I obtained a Minimal Experimental Arm only recoding (xi) = 14 and a Minimal Control Arm only recoding (yi) = 15, whereas the authors reported that 5 recodings were sufficient. It appears that differences in code versions or functions might have slightly affected the results. After downgrading R and running the eoic function with PlaneGeometry successfully installed, the fragility index for the EOI approach was 15 rather than 5.*

Apologies for the issue with PlaneGeometry, I will try to fix this for future iterations. The difference you see is an artefact of running EOIFUNC on pooled data, rather than the dedicated EOIMETA function, with the chief difference being that EOIFUNC doesn't apply WIV correction. If we simply pool events, this is the output:

```
**Total Sample:** 133262
**p-value:** 0.09796542
**Hazard Ratio (Exp/Con):** 0.9340521

**FECK UP Vector co-ordinates:**
  X-point: 7.448981
  Y-point: 7.097156
  FECKUP vector length 10.28868
  Resolved FECKUP length (dmin)  15 subjects

**Single arm re-coding summary:**
  Minimal Exp Arm only recoding (xi) : 14.21051
  Minimal Con Arm only recoding (yi) : 14.91487
```

Author response image 1.

If the reviewer uses the EOIMETA function which employs inverse weighing, then to define each trial we use a vector of events and non-events in each arm. For all the 12 studies, this would be (in R code syntax, or import from github file)

```
av <- c(63,344,151, 0,1, 51,12,30,58,154,14,242) #exp events

bv <- c(1282,17832,2498,91,121,17360, 65,2514,193,12143,788,10419) # exp
non-evs

cv <- c(72,382,178,1,1,66,12,30,51,187,5,210)  #con events

dv <- c(1269,17724,2465,90,117,17286,66,2505,115,12757,808,10439) #con non-
evs

eoimeta(av,bv,cv,dv) #run EOIMETA with weighed inverse

#use eoimeta(av,bv,cv,dv, atal = 1) for atal fragility metric (slower)
```

Author response image 2.

Then they will obtain:

```
**Meta Summary:** 133262
**p-value (cmh):** 0.09972119
**p-value (crude):** 0.06104107
**RR random:** 0.9159249
**CMH Hazard Ratio (Exp/Con):** 0.9248945 0.9159249
**Crude Hazard Ratio (Exp/Con):** 0.9253417
**95% Confidence bounds (Exp/Con):** 0.8249934 1.016879
**Confounding (%):** 1.017655

**Low degree of confounding, grouping is possible:**

**Crude pooled EOI / ROAR figures:**
  Total events (Experimental arm) : 1099
  Total non-events (Experimental arm) : 65327
  Total events (Control arm) : 1195
  Total non-events (Control arm) : 65641

**EOI Fractions:**
  Experimental arm re-coding fraction (%): 0.006062537
  Control arm re-coding fraction (%): 0.006309097
  Total sample re-coding fraction (%): 0.003752007

**ROAR Fractions:**
  Experimental arm re-coding fraction (%): 0.2831601
  Control arm re-coding fraction (%): 100
  Total sample re-coding fraction (%): 0.1438698

**Summary stats:**
  Resolved FECKUP vector (EOI, dmin): 5
  Resolved FOCK vector (ROAR, rmin): 192

**Atal's method fragility index:**
  Atal fragility: NA
> #use eoimeta(av,bv,cv,dv, atal = 1) for atal fragility metric (slower)
```

Author response image 3.

If the reviewer runs a simple pooler analysis with weighed inverse correction turned off, they should return a similar answer as a simple eoifunc call, save the zero count correction difference. But EOIMETA weighs the sample, and is reported in main paper.

> *(12) I recalculated the eoic function for Zhang et al. (2019) and found a fragility index (dmin) of 1. FECKUP Vector Length: 0.5722. Minimal Experimental Arm Recoding (xi): 0.7738. Minimal Control Arm Recoding (yi): 0.8499.*

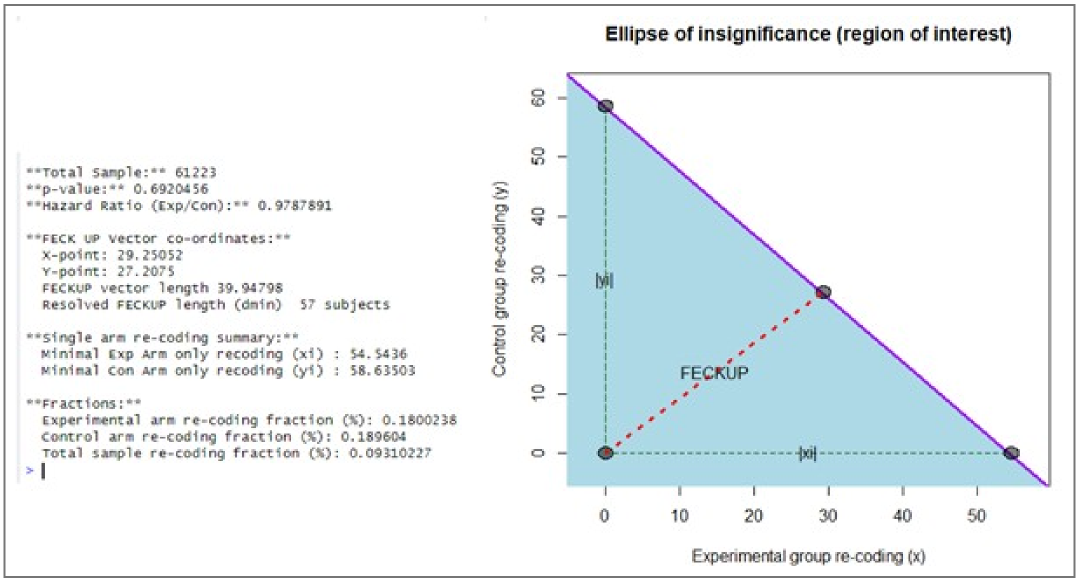
This again appears to be an artefact of using eoifunc rather than eoimeta; with eoimeta, which uses WIV to adjust the studies for heterogeneity effects, this is the reported output:

```
**Meta Summary:** 38538
**p-value (cmh):** 0.03494057
**p-value (crude):** 0.03858413
**RR random:** 0.8680809
**CMH Hazard Ratio (Exp/Con):** 0.8684754 0.8680809
**Crude Hazard Ratio (Exp/Con):** 0.8695922
**95% Confidence bounds (Exp/Con):** 0.7614684 0.989620
**Confounding (%):** 0.1737895

**Low degree of confounding, grouping is possible:**

**Crude pooled EOI / ROAR figures:**
  Total events (Experimental arm) : 394
  Total non-events (Experimental arm) : 18563
  Total events (Control arm) : 468
  Total non-events (Control arm) : 19113

**EOI Fractions:**
  Experimental arm re-coding fraction (%): 0.01586494
  Control arm re-coding fraction (%): 0.01692308
  Total sample re-coding fraction (%): 0.01037937

**ROAR Fractions:**
  Experimental arm re-coding fraction (%): 0.0162145
  Control arm re-coding fraction (%): 0.7294512
  Total sample re-coding fraction (%): 0.007783918

**Summary stats:**
  Resolved FECKUP vector (EOI, dmin): 4
  Resolved FOCK vector (ROAR, rmin): 3

**Atal's method fragility index:**
  Atal fragility: NA
```

Author response image 4.

> *(13) Using the previous code (before downgrading R and loading PlaneGeometry), I recalculated the EOI for Zhang et al. (2022) and found Minimal Experimental Arm only recoding (xi) = 55 and Minimal Control Arm only recoding (yi) = 59-results slightly closer to those reported by the authors. After properly loading PlaneGeometry, I recalculated and obtained for Zhang et al. (2022): Fragility index (dmin) = 57; FECKUP Vector Length = 39.948; Minimal Experimental Arm Recoding (xi) = 54.5436; Minimal Control Arm Recoding (yi) = 58.635.*

Again this appears to be a difference in using eoifunc or eoimeta as a call - I can replicate this result using EOIFUNC:



Author response image 5.

But adjusting for study weighing with eoimeta:

```
**Summary stats:**
  Resolved FECKUP vector (EOI, dmin): 51
  Resolved FOCK vector (ROAR, rmin): 96
```

Author response image 6.

> *(14) For Guo et al. (2022), the EOI fragility index was 17 [dmin = 17]. FECKUP Vector Length: 11.3721. Minimal Experimental Arm Recoding (xi): -15.6825. Minimal Control Arm Recoding (yi): -16.5167. However, the authors report an EOI fragility of 38. Since I was able to load PlaneGeometry properly and run eoicfunc.R (from GitHub) without errors, the discrepancies likely reflect minor coding or version inconsistencies rather than software limitations.*

These again stem from using eoifunc on simple pooled data versus eoimeta, which adjusts by study.

> *(15) Replications of ROAR fragility: roarfunc.R (github): - For Guo et al. (2022), the ROAR fragility calculated using roarfunc.R was 16 [rmin (Redaction Fragility Index) = 16]. FOCK Vector Length: 15.942. Minimal Experimental Arm Redaction (xc): 15.9442. Minimal Control Arm Redaction (yc): 978.8906. In the main text, the author reports a redaction fragility of 37. What might explain these discrepancies?*

Again, this stems from EOIMETA versus EOIFUNC (and roarfunc calls without weighed adjustment). As the reviewer has observed, the fragility increases when there is no study level adjustment, which we have now added to the discussion text.

> *(16) In generic_run.R, line 6 contains a bug - it is missing a forward slash (/) between the directory path and the filename. The correct line of code should be: pathload = paste0(pathname, "/", filename, exname). The same issue occurs in generalcode.R.*

Apologies, I will correct this in the upload!

> *(17) Theoretical framework: Is there any other method available for comparison besides the one proposed by Atal et al.? Could you include a brief literature review describing alternative approaches?*

To my knowledge, there is not – Xing et al (now referenced) covered this earlier in the year, and I have included an expanded background for this purpose. Please see reply to reviewer 1, point 1.

> *(18a) There appears to be no heterogeneity in the meta-analysis in terms of effect sizes and I^2, likely because most values are quite large, yet the included studies address very different populations (e.g., patients with COPD, NSCLC survivors, older adults, women, and GI cancer survivors). This could have been explained more clearly, including how such diverse literature might influence fragility indices or whether there is a logical rationale for combining these studies. Could you perform a sensitivity analysis or provide a conceptual explanation of how the heterogeneity - or lack thereof - across these trials may affect the fragility indices? Although I^2 values are small, the conceptual heterogeneity among studies suggests that the pooled results may be comparing fundamentally different clinical contexts, which requires clarification.*

I think this is a very pertinent point, I am unsure as to why these authors combined such diverse populations without any consideration of whether they were comparable, but this is a common problem in meta-analysis. I have added the following to the discussion to address this problem:

"The use of vitamin D meta-analyses in this work was chosen as illustrative rather than specific, but it is worth noting that there are methodological concerns with much vitamin D research. (Grimes aet al., 2024). The three studies cited in this work report relatively low heterogeneity in their meta-analysis in both effect sizes and I^2 values, but it is worth noting that the included studies addressed very different populations, including patients with Chronic Obstructive Pulmonary Disease, Non small cell lung cancer survivors, women only cohorts, older adults, and gastrological cancer survivors. These groups have presumably different risk factors for cancer deaths, and why the authors of these studies combined the cohorts with fundamentally different clinical contexts is unclear. Why the heterogeneity appeared so relatively low in different groups is also a curious feature. This goes beyond the scope of the current work, but serves as an example of the reality that meta-analysis is only as strong as its underlying data and methodological rigor in comparing like-with-like, and the conclusions drawn from them must always be seen in context."

> ***Reviewer #3 (Recommendations for the authors):***
>
> *(1) Line 156, acronym FI not defined.*

Apologies, I this is now defined at the outset as "fragility index".

> *(2) Line 158, typo "He"?*

Apologies again, this was a typo and was supposed to read "the", fixed now.

> *(3) Across the manuscript, I think the "re-coding" phrasing may confuse clinical readers. Maybe rephrasing to "flipping event classification" or "flipping group" would be better.*

Excellent point – this has now been modified at the outset.

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